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COURSE NAME : ANALOG AND DIGITAL COMMUNICATION (ECA0802)

|           |                                                                             |
|-----------|-----------------------------------------------------------------------------|
| EX NO: 2a | GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS<br>USING MATLAB |
| DATE:     |                                                                             |

## Objectives

- (i) To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.
- (ii) To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

## Objective 1:

To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.

## PROGRAM (MATLAB CODE)

```
clc;  
  
clear;  
  
close all;  
  
Am=1;  
  
Ac=1;  
  
fm=5;  
  
fc=50;  
  
beta=5;  
  
kp=5;  
  
fs=1000;
```

```

t=0:1/fs:1;

m = Am*cos(2*pi*fm*t);           % Message signal

c = Ac*cos(2*pi*fc*t);           % Carrier signal

fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM

subplot(4,1,1), plot(t,m), title('Message Signal')

subplot(4,1,2), plot(t,c), title('Carrier Signal')

subplot(4,1,3), plot(t,fm_sig), title('FM Signal')

```

### Procedure:

1. Open MATLAB and create a new script file.
2. Clear the MATLAB workspace, command window, and close all existing figure windows using:

```
clc; clear; close all;
```

3. Define the message signal parameters:
  - Message amplitude  $A_m = 1$
  - Message frequency  $f_m = 5\text{Hz}$
4. Define the carrier signal parameters:
  - Carrier amplitude  $A_c = 1$
  - Carrier frequency  $f_c = 50\text{Hz}$
5. Set the modulation index  $\beta = 5$  and sampling frequency  $f_s = 1000\text{Hz}$ .
6. Generate the time vector from 0 to 1 second with a sampling interval of  $1/f_s$ :

```
t = 0:1/fs:1;
```

7. Generate the message signal using:

```
m = Am*cos(2*pi*fm*t);
```

This produces a low-frequency sinusoidal signal that acts as the modulating signal.

8. Generate the carrier signal using:

```
c = Ac*cos(2*pi*fc*t);
```

This produces a high-frequency sinusoidal carrier.

9. Generate the Frequency Modulated (FM) signal using:

```
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
```

Here, the instantaneous frequency of the carrier varies according to the amplitude of the message signal.

10. Plot the message signal using:

```
subplot(4,1,1), plot(t,m), title('Message Signal')
```

11. Plot the carrier signal using:

```
subplot(4,1,2), plot(t,c), title('Carrier Signal')
```

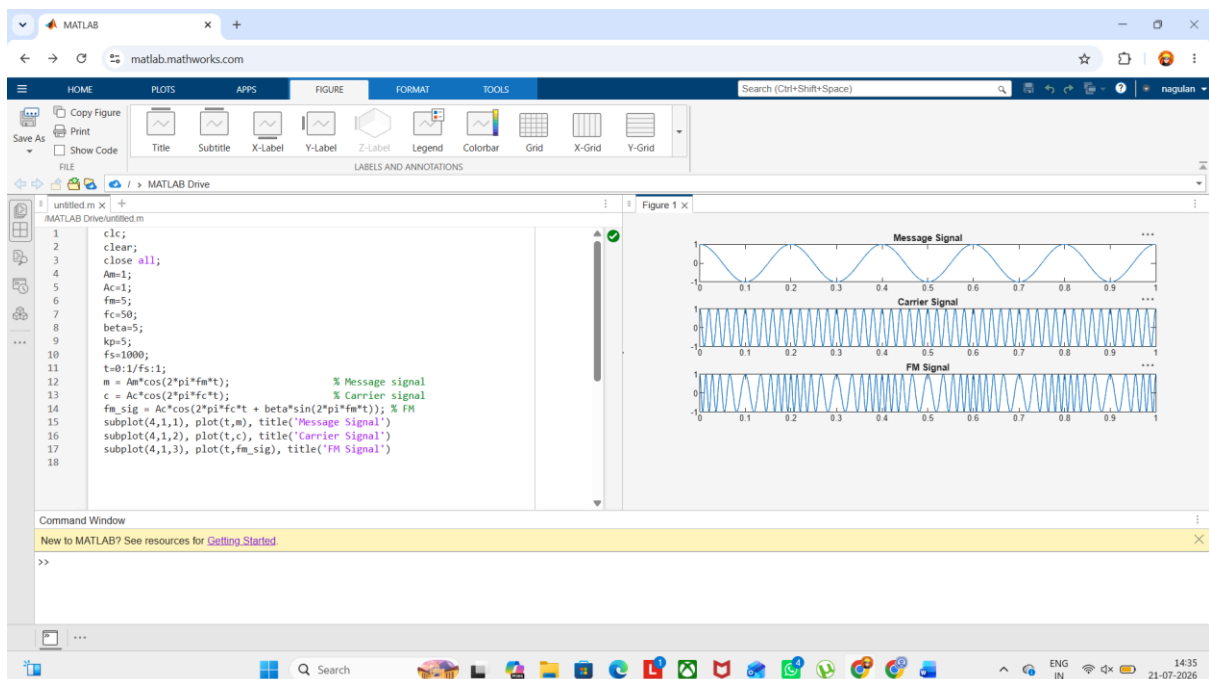
12. Plot the generated FM signal using:

```
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
```

13. Execute the program and observe the waveforms of the message signal, carrier signal, and FM signal.

14. Analyze the FM waveform and note that the frequency of the carrier varies according to the amplitude of the message signal while the amplitude of the FM signal remains constant.

## OUTPUT (GRAPH)



## Result

The Frequency Modulated (FM) signal was successfully generated using MATLAB, and the time-domain waveforms of the message signal, carrier signal, and FM signal were observed and analyzed.

## Objective 2:

To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule

## PROGRAM (MATLAB CODE)

```
% Calculation and Verification of Frequency Deviation
```

```
% and Modulation Index of FM Signals
```

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Input Parameters
```

```
Am = 2;          % Message signal amplitude
```

```
fm = 500;        % Message frequency (Hz)
```

```
Ac = 1;          % Carrier amplitude
```

```
fc = 10000;      % Carrier frequency (Hz)
```

```
kf = 1000;       % Frequency sensitivity (Hz/V)
```

```
fs = 100000;     % Sampling frequency (Hz)
```

```
T = 0.02;        % Simulation duration (s)
```

```
t = 0:1/fs:T;
```

```
%% Message Signal
```

```
m = Am*cos(2*pi*fm*t);
```

```
%% Frequency Deviation Calculation
```

```
delta_f = kf * Am;
```

```
%% Modulation Index Calculation
```

```

beta = delta_f/fm;

%% FM Signal Generation
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Plot Message Signal
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% Plot FM Signal
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% Spectrum Analysis
N = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal))/N);
f = (-N/2:N/2-1)*(fs/N);

figure;
plot(f,FM_fft,'LineWidth',1.5);
title('FM Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

```

```

xlim([fc-10000 fc+10000]);

%% Display Results

fprintf('\nFM SIGNAL PARAMETERS\n');

fprintf('-----\n');

fprintf('Carrier Frequency    = %.0f Hz\n',fc);

fprintf('Message Frequency     = %.0f Hz\n',fm);

fprintf('Message Amplitude      = %.2f V\n',Am);

fprintf('Frequency Sensitivity = %.0f Hz/V\n',kf);

fprintf('\nCALCULATED VALUES\n');

fprintf('-----\n');

fprintf('Frequency Deviation ( $\Delta f$ ) = %.0f Hz\n',delta_f);

fprintf('Modulation Index ( $\beta$ )    = %.2f\n',beta);

%% Carson's Rule Bandwidth

BW = 2*(delta_f + fm);

fprintf('Bandwidth (Carson Rule) = %.0f Hz\n',BW);

```

## Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all existing figure windows using:

```
clc;
```

```
Clear;
```

```
close all;
```

3. Define the message signal parameters:

- Message amplitude  $A_m=2V$
- Message frequency  $f_m=500Hz$

4. Define the carrier signal parameters:

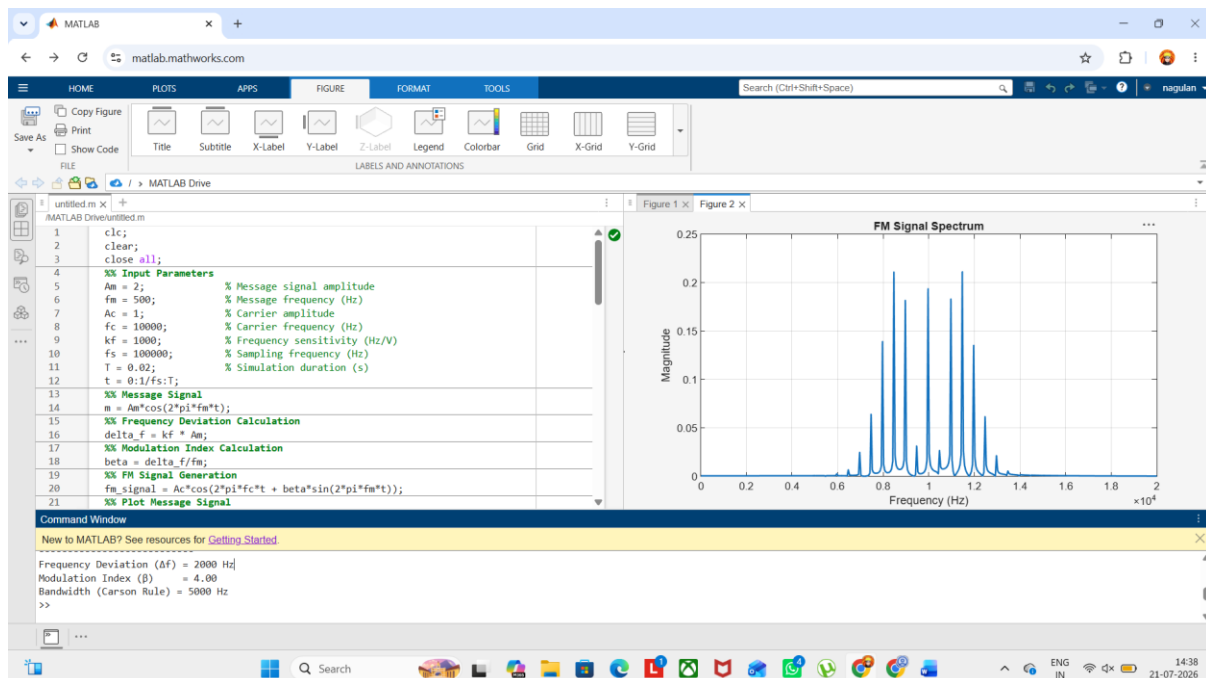
- Carrier amplitude  $A_c=1V$

- Carrier frequency  $f_c=10000\text{Hz}$
5. Set the frequency sensitivity constant  $k_f=1000\text{Hz/V}$ , sampling frequency  $f_s=100000\text{Hz}$ , and simulation duration  $T=0.02\text{s}$ .
  6. Generate the time vector using:
 
$$t = 0:1/f_s:T;$$
  7. Generate the message signal using:
 
$$m = A_m \cos(2\pi f_m t);$$
  8. Calculate the frequency deviation of the FM signal using:
 
$$\Delta f = k_f A_m$$
  9. Calculate the modulation index using:
 
$$\beta = \Delta f / f_m$$
  10. Generate the FM signal using:
 
$$fm\_signal = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t));$$
  11. Plot the message signal in the time domain and observe its waveform.
  12. Plot the generated FM signal and observe that the carrier frequency varies according to the message signal while the amplitude remains constant.
  13. Determine the length of the FM signal:
 
$$N = \text{length}(fm\_signal);$$
  14. Compute the Fast Fourier Transform (FFT) of the FM signal:
 
$$FM\_fft = \text{fft}(fm\_signal);$$
  15. Shift the spectrum to center the zero-frequency component and normalize it:
 
$$FM\_fft = \text{fftshift}(\text{abs}(FM\_fft)/N);$$
  16. Generate the frequency axis corresponding to the FFT output:
 
$$f = (-N/2:N/2-1)*(f_s/N);$$
  17. Plot the frequency spectrum of the FM signal and observe the carrier component and sidebands.
  18. Restrict the frequency axis around the carrier frequency for better visualization:
 
$$\text{xlim}([f_c-10000 \ f_c+10000]);$$
  19. Calculate the bandwidth of the FM signal using Carson's Rule:

$$BW=2(\Delta f+f_m)$$

20. Display the carrier frequency, message frequency, frequency deviation, modulation index, and bandwidth in the MATLAB command window.
21. Execute the program and verify that the calculated frequency deviation and modulation index match the values used in generating the FM signal

## OUTPUT:



## Result

- i) The time-domain waveform was observed, and the frequency spectrum obtained using FFT has been plotted.
- ii) Also, calculated the frequency deviation and bandwidth using Carson's rule.



